

Collaborative Competition in Offshore Capital Networks

Soumadeep Ghosh

Kolkata, India

Abstract

We develop a comprehensive formal theory of *collaborative competition* in offshore capital networks, wherein speculative investors from n nations $\{N_1, \dots, N_n\}$ interact with h tax havens $\{H_1, \dots, H_h\}$ as both collaborators and rivals. We introduce the *Duality Ratio* Δ_{ijk} to measure the balance between cooperative and competitive forces across investor–haven triads, and characterise four macro-regimes—Cartel Equilibrium, Fragmented Rivalry, Captured Haven, and Network Flux—under which distinct equilibria emerge. Formal results include: a multi-level game-theoretic formulation; a stochastic threshold theorem for regulatory dissolution; a Laplacian network model with spectral stability analysis; a Bayesian hierarchical framework for empirical identification; and a graph-neural-network (GNN) algorithm for real-time regime classification. We further prove the *Identity Paradox*: each investor simultaneously requires rivals for the differential rents that make offshore access valuable, and collaborators to sustain the institutional infrastructure that provides that access. Simulations and stylised data confirm the theoretical predictions.

The paper ends with “The End”

Keywords: tax havens, offshore capital, collaborative competition, game theory, network economics, regulatory threshold, graph neural networks.

Contents

1	Introduction	2
2	The Formal Model	3
2.1	Primitives and Notation	3
2.2	The Duality Ratio	3
2.3	The Duality Tensor	3
2.4	Capital Flows and Return Functions	4
3	Game-Theoretic Analysis	4
3.1	The Multi-Level Game	4
3.2	Nash Equilibria and Regime Classification	4
3.3	The Regulatory Paradox Theorem	5
4	Network and Spectral Theory	5
4.1	Bipartite Network Representation	5
4.2	Graph Laplacian and Spectral Stability	5
4.3	Haven Centrality	6
5	Statistical and Econometric Framework	6
5.1	Bayesian Hierarchical Model	6
5.2	Regime Identification via Latent Class Model	6
5.3	EM Estimation	7
5.4	Testing for Structural Breaks	7
6	AI/ML: Graph Neural Network Regime Classifier	7
6.1	GNN Architecture	7
6.2	CCOCN-GNN Regime Classifier	8

7	The Identity Paradox	9
8	Simulations	9
8.1	Phase Diagram	9
8.2	Bipartite Network Topology	10
8.3	Regulatory Threshold Simulation	10
9	Implications and Conclusion	11
9.1	Cross-Disciplinary Implications	11
9.2	Policy Recommendations	11
9.3	Conclusion	11

List of Figures

1	Phase diagram of CCOCN macro-regimes. Dashed lines mark critical thresholds $\bar{\Delta} = 1$ and $\lambda_2 = 0.5$; coloured dots are individual Monte Carlo draws.	9
2	CCOCN bipartite graph: $n = 5$ investors and $h = 3$ havens under Cartel Equilibrium. Thick blue edges denote dominant capital flows; H_2 attains the highest betweenness centrality (Theorem 4.3).	10
3	Simulated trajectory of mean duality $\bar{\Delta}(\theta)$. Collaboration is fortified below θ^* ; the network dissolves beyond $\tilde{\theta}$. The dotted line marks $\bar{\Delta} = 1$	10

List of Tables

1	Four Macro-Regimes of the CCOCN Game	4
2	Stylised Summary Statistics by Regime (5,000-iteration Monte Carlo)	7
3	CCOCN-GNN Performance (5-fold Cross-Validation, $n \in [5, 50]$, $h \in [2, 20]$)	8
4	Cross-Disciplinary Implications of the CCOCN Framework	11

1 Introduction

The global offshore financial system constitutes one of the most structurally complex networks in modern political economy. Approximately \$8–12 trillion in private wealth is held offshore [Zucman \(2013\)](#), channelled through a small number of jurisdictions—the Cayman Islands, Luxembourg, Singapore, and Delaware, among others—that collectively form the architectural backbone of international tax arbitrage.

What makes this system theoretically remarkable is not merely its scale but its *relational structure*. Speculative investors from different nations simultaneously cooperate and compete: they collaborate to sustain the political and legal viability of haven jurisdictions (a public good), yet fiercely rival one another for privileged access to the differential rents those havens produce (a private good). Standard game-theoretic frameworks, which typically model actors as either purely cooperative or purely competitive, cannot adequately capture this duality.

This paper constructs a unified formal theory—which we term **Collaborative Competition in Offshore Capital Networks (CCOCN)**—synthesising tools from game theory, financial economics, network science, Bayesian statistics, and machine learning. Our contributions are as follows.

- (i) A rigorous definition of the *Duality Ratio* Δ_{ijk} and its associated network tensor.
- (ii) A multi-level game whose Nash equilibria map exactly onto four empirically observable macro-regimes.
- (iii) A regulatory threshold theorem establishing conditions for network dissolution, with a proof that moderate regulation is self-defeating.
- (iv) A spectral stability analysis of the haven network using graph Laplacians.
- (v) A Bayesian hierarchical model for empirical identification of regimes from financial-flow data.
- (vi) A graph-neural-network algorithm for real-time regime classification.

The paper is organised as follows. Section 2 lays out the formal model. Section 3 develops the game-theoretic analysis. Section 4 presents network and spectral theory. Section 5 provides the statistical framework. Section 6 develops the machine-learning algorithm. Section 7 proves the Identity Paradox. Section 8 presents simulations. Section 9 concludes.

2 The Formal Model

2.1 Primitives and Notation

Let $\mathcal{N} = \{N_1, \dots, N_n\}$ be a finite set of *speculative investors* indexed by nation of domicile, and let $\mathcal{H} = \{H_1, \dots, H_h\}$ be a finite set of *tax haven jurisdictions*.

Definition 2.1 (Investor Profile). Investor N_i is characterised by the tuple

$$N_i = (\kappa_i, \rho_i, \pi_i, \lambda_i),$$

where $\kappa_i \in \mathbb{R}_{>0}$ is capital endowment, $\rho_i \in [0, 1]$ is political influence capacity, $\pi_i: \mathcal{H} \rightarrow \mathbb{R}$ is a preference ordering over havens, and $\lambda_i \in (0, 1)$ is a discount factor.

Definition 2.2 (Haven Profile). Haven H_j is characterised by the tuple

$$H_j = (\sigma_j, \alpha_j, \phi_j),$$

where $\sigma_j \in [0, 1]$ measures *opacity supply* (secrecy provision), $\alpha_j \in \mathbb{R}_{>0}^n$ is the vector of *preferential allocations* to each investor, and $\phi_j \in [0, 1]$ is *switching leverage*—the credible capacity of H_j to reallocate preferences among investors.

2.2 The Duality Ratio

For any investor pair (N_i, N_j) with $i \neq j$ and any haven H_k , define the scalar functions

$$C(N_i, N_j, H_k) = \sigma_k \cdot (\rho_i + \rho_j) \cdot \log(1 + \kappa_i \kappa_j), \quad (1)$$

$$R(N_i, N_j, H_k) = \frac{\alpha_k(i) - \alpha_k(j)}{\alpha_k(i) + \alpha_k(j)} \cdot \phi_k \cdot (\kappa_i + \kappa_j), \quad (2)$$

where C measures the *collaboration value* (joint gain from sustaining H_k) and R measures the *rivalry cost* (loss from competing for H_k 's differential rents).

Definition 2.3 (Duality Ratio). The **Duality Ratio** of triad (N_i, N_j, H_k) is

$$\Delta_{ijk} := \frac{C(N_i, N_j, H_k)}{R(N_i, N_j, H_k)} \in \mathbb{R}_{>0}, \quad \forall i \neq j, \quad \forall k.$$

Assumption 2.4 (Strict Positivity). For all $i \neq j$ and all k : $C(N_i, N_j, H_k) > 0$ and $R(N_i, N_j, H_k) > 0$, so that $\Delta_{ijk} \in (0, \infty)$.

Assumption 2.4 encodes the *structural duality axiom*: the network is never purely cooperative nor purely competitive.

2.3 The Duality Tensor

Collect all pairwise ratios into an order-3 tensor $\Delta \in \mathbb{R}^{n \times n \times h}$ with $[\Delta]_{ijk} = \Delta_{ijk}$ for $i \neq j$ and $[\Delta]_{iik} = 0$. The *mean duality* is

$$\bar{\Delta} = \frac{1}{n(n-1)h} \sum_{\substack{i,j=1 \\ i \neq j}}^n \sum_{k=1}^h \Delta_{ijk}.$$

When $\bar{\Delta} > 1$ collaboration dominates in aggregate; when $\bar{\Delta} < 1$ rivalry dominates.

2.4 Capital Flows and Return Functions

Let $x_{ij} \in [0, \kappa_i]$ denote capital routed by N_i through H_j . The *after-tax return* to N_i via H_j is

$$r_{ij}(x_{ij}, \mathbf{x}_{-i,j}) = \mu_{ij} - \tau_j(X_j) \cdot x_{ij} + \alpha_j(i) \cdot \sigma_j - \phi_j \cdot \frac{x_{ij}}{X_j},$$

where μ_{ij} is the pre-tax return, $\tau_j(X_j)$ is the effective tax rate (non-decreasing in aggregate flow $X_j = \sum_i x_{ij}$), and the final term captures the *crowding cost* of competition. Total investor payoff is

$$\Pi_i(\mathbf{x}_i, \mathbf{x}_{-i}) = \sum_{j=1}^h r_{ij}(x_{ij}, \mathbf{x}_{-i,j}) - c_i(\mathbf{x}_i),$$

where $c_i: \mathbb{R}^h \rightarrow \mathbb{R}_{\geq 0}$ is a convex routing cost function.

3 Game-Theoretic Analysis

3.1 The Multi-Level Game

The interaction is modelled as a two-tier simultaneous game.

Level 1 (Haven Sustainability Game). Investors choose lobbying efforts $\ell_i \in [0, \bar{\ell}]$, minimising the probability that haven H_k collapses:

$$\Pr[H_k \text{ collapses}] = \exp\left(-\gamma_k \sum_{i=1}^n \ell_i\right),$$

where $\gamma_k > 0$ is haven k 's fragility parameter. This is a voluntary-contribution public-good game.

Level 2 (Rent-Capture Game). Conditional on haven survival, investors compete for preferential allocation $\alpha_k(i)$ through capital commitment x_{ik} :

$$\alpha_k(i) = \frac{x_{ik}^\beta}{\sum_{j=1}^n x_{jk}^\beta}, \quad \beta > 0,$$

a *Tullock contest success function* [Tullock \(1980\)](#) with elasticity β .

3.2 Nash Equilibria and Regime Classification

Definition 3.1 (Macro-Regime). A **macro-regime** $\mathcal{R}$ is a partition of the joint strategy space characterised by the distribution of $\{\Delta_{ijk}\}$ and the number of active havens $h^* \leq h$.

Theorem 3.2 (Four-Regime Characterisation). *Under Assumption 2.4, the CCOCN game admits exactly four qualitatively distinct Nash equilibrium classes, described in Table 1.*

Proof. The joint strategy space $\mathcal{S} = [0, \bar{\ell}]^n \times \prod_{i=1}^n [0, \kappa_i]^h$ is compact and convex, and each payoff Π_i is continuous in all strategies. By Kakutani's Fixed-Point Theorem [Kakutani \(1941\)](#) a Nash equilibrium exists. Equilibria are partition-classified by two binary conditions: (i) $\bar{\Delta} \geq 1$ and (ii) $h^* \geq h/2$. The four cells of this partition correspond exactly to the four regimes in Table 1. $\square$ $\square$

Table 1: Four Macro-Regimes of the CCOCN Game

Regime	$\bar{\Delta}$	h^*	Power Locus	Characteristic Behaviour
Cartel Equilibrium	> 1	Low	Investors	Investors collude; havens compete for inflows
Fragmented Rivalry	< 1	High	Havens	Investors defect; havens gain bargaining power
Captured Haven	$\gg 1$	1	N_i^*	One investor dominates; haven loses pivot status
Network Flux	≈ 1	Any	Unstable	High shock sensitivity; bifurcation-prone

3.3 The Regulatory Paradox Theorem

Let $\theta \in [0, 1]$ denote *regulatory pressure* exerted by non-haven states ($\theta = 0$: no enforcement; $\theta = 1$: complete elimination).

Theorem 3.3 (Regulatory Paradox). *There exists a critical threshold $\theta^* \in (0, 1)$ such that:*

- (i) for $\theta < \theta^*$: $\frac{\partial \bar{\Delta}}{\partial \theta} > 0$ (pressure strengthens collaboration);
- (ii) at $\theta = \theta^*$: $\bar{\Delta}$ attains its global maximum;
- (iii) for $\theta > \theta^*$: $\frac{\partial \bar{\Delta}}{\partial \theta} < 0$ and the network dissolves as $\theta \rightarrow 1$.

Proof. Regulatory pressure enters through two channels.

Channel 1 (Cost effect). Higher θ raises effective access costs: $\tau_j \mapsto \tau_j(1 + \eta_j\theta)$, $\eta_j > 0$. This shrinks differential rents, reduces R , and raises Δ_{ijk} —a collaboration-strengthening effect.

Channel 2 (Lobbying effect). Higher θ raises the marginal benefit of joint lobbying, increasing the equilibrium effort ℓ_i^* for every i , further reinforcing collaboration.

Both channels increase $\bar{\Delta}$ for $\theta < \theta^*$. For $\theta > \theta^*$, however, the expected profit from haven use falls below the total coordination cost. Formally, define

$$V(\theta) = \mathbb{E}[\Pi_i^*(\theta)] - \mathbb{E}[c_i(\ell_i^*(\theta))].$$

By the Envelope Theorem and convexity of c_i , V is unimodal in θ with unique maximiser θ^* satisfying $V'(\theta^*) = 0$. Network dissolution occurs when $V(\theta) < 0$, which holds for all $\theta > \tilde{\theta} > \theta^*$. The global maximum of $\bar{\Delta}$ coincides with θ^* by continuity. $\square$ $\square$

Corollary 3.4. *Moderate multilateral regulation—such as the OECD Pillar Two global minimum tax with weak enforcement ($\theta < \theta^*$)—is not merely ineffective but counterproductive: it fortifies the CCOCN network by raising $\bar{\Delta}$.*

4 Network and Spectral Theory

4.1 Bipartite Network Representation

The CCOCN system is represented as a weighted bipartite graph $\mathcal{G} = (\mathcal{N} \cup \mathcal{H}, \mathcal{E}, w)$, where edge weight $w(N_i, H_k) = x_{ik}$ is the capital flow. The weighted adjacency matrix satisfies

$$\mathbf{A} \in \mathbb{R}^{(n+h) \times (n+h)}, \quad A_{ij} = \begin{cases} x_{ij} & \text{if } (N_i, H_j) \in \mathcal{E}, \\ 0 & \text{otherwise,} \end{cases}$$

and, because $\mathcal{G}$ is bipartite, takes the block form

$$\mathbf{A} = \begin{pmatrix} \mathbf{0}_{n \times n} & \mathbf{X} \\ \mathbf{X}^\top & \mathbf{0}_{h \times h} \end{pmatrix}, \quad \mathbf{X} \in \mathbb{R}^{n \times h}.$$

4.2 Graph Laplacian and Spectral Stability

Let $\mathbf{D}$ be the diagonal degree matrix of $\mathcal{G}$. The *normalised graph Laplacian* is

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2},$$

with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{n+h} \leq 2$.

Definition 4.1 (Algebraic Connectivity). The **algebraic connectivity** (Fiedler value) of the CCOCN network is $\lambda_2(\mathbf{L})$. Larger values indicate greater cohesion.

Proposition 4.2 (Spectral Regime Signatures). *The four regimes of Theorem 3.2 exhibit the following spectral signatures:*

Cartel Equilibrium: λ_2 large; spectrum concentrated near 1;
Fragmented Rivalry: λ_2 small; spectrum dispersed;
Captured Haven: $\lambda_2 \approx 0$ (network near-disconnected);
Network Flux: λ_2 volatile under perturbation.

Proof. In the Cartel regime all x_{ik} are large and approximately equal, yielding a near-regular bipartite graph whose Laplacian spectrum clusters around eigenvalue 1 Chung (1997). In Fragmented Rivalry flows concentrate on few edges, dispersing the spectrum. In Captured Haven the dominant investor–haven subgraph is nearly disconnected, so $\lambda_2 \approx 0$ by Cheeger’s inequality. Network Flux corresponds to **A** near a bifurcation point where small perturbations in **X** cause large swings in λ_2 . $\square$ $\square$

4.3 Haven Centrality

Let p_{ij} denote the number of shortest paths from N_i to N_j in $\mathcal{G}$, and $p_{ij}(H_k)$ the number of those paths passing through haven H_k . The *betweenness centrality* of H_k is

$$B(H_k) = \sum_{\substack{i,j \in \mathcal{N} \\ i \neq j}} \frac{p_{ij}(H_k)}{p_{ij}}.$$

Theorem 4.3 (Haven Dominance in Centrality). *In any CCOCN equilibrium, $\max_k B(H_k) > \max_i B(N_i)$.*

Proof. In the bipartite graph $\mathcal{G}$ no direct N_i – N_j edges exist, so every path between two investor nodes must pass through at least one H_k . Hence $p_{ij}(H_k) > 0$ for every active haven and every investor pair, and summing over all $\binom{n}{2}$ pairs gives haven betweenness strictly greater than that of any investor node. $\square$ $\square$

5 Statistical and Econometric Framework

5.1 Bayesian Hierarchical Model

Let $\mathbf{F} \in \mathbb{R}^{n \times h}$ be the observed capital-flow matrix (a noisy proxy for **X**). We specify the hierarchical model

$$F_{ij} \mid x_{ij}, \sigma_\varepsilon^2 \sim \mathcal{N}(x_{ij}, \sigma_\varepsilon^2), \quad (3)$$

$$x_{ij} \mid \mu_j, \tau_j^2 \sim \mathcal{N}^+(\mu_j, \tau_j^2), \quad (4)$$

$$\mu_j \sim \mathcal{N}(\mu_0, s_0^2), \quad (5)$$

$$\tau_j^2 \sim \text{Inv-Gamma}(a, b), \quad (6)$$

$$\sigma_\varepsilon^2 \sim \text{Inv-Gamma}(a_\varepsilon, b_\varepsilon), \quad (7)$$

where $\mathcal{N}^+$ denotes the half-normal (flows are non-negative). Posterior inference proceeds via No-U-Turn Sampling Hoffman & Gelman (2014).

5.2 Regime Identification via Latent Class Model

Regime membership is a latent categorical variable $Z \in \{1, 2, 3, 4\}$. We posit

$$Z \mid \boldsymbol{\pi} \sim \text{Categorical}(\boldsymbol{\pi}), \quad \boldsymbol{\pi} \sim \text{Dir}(\boldsymbol{\alpha}_0),$$

with the bivariate Gaussian likelihood

$$p(\hat{\lambda}_2, \hat{\Delta} \mid Z = r) = \mathcal{N}_2\left(\begin{pmatrix} \hat{\lambda}_2 \\ \hat{\Delta} \end{pmatrix} \mid \boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r\right),$$

where $(\boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r)$ are regime-specific Gaussian parameters estimated by expectation-maximisation.

5.3 EM Estimation

E-step. Compute posterior regime responsibilities:

$$\gamma_{tr} = \frac{\pi_r \mathcal{N}_2(\mathbf{y}_t \mid \boldsymbol{\mu}_r, \boldsymbol{\Sigma}_r)}{\sum_{s=1}^4 \pi_s \mathcal{N}_2(\mathbf{y}_t \mid \boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)}, \quad \mathbf{y}_t = (\hat{\lambda}_{2,t}, \hat{\Delta}_t)^\top.$$

M-step. Update parameters:

$$\begin{aligned} \hat{\pi}_r &= \frac{1}{T} \sum_{t=1}^T \gamma_{tr}, \\ \hat{\boldsymbol{\mu}}_r &= \frac{\sum_{t=1}^T \gamma_{tr} \mathbf{y}_t}{\sum_{t=1}^T \gamma_{tr}}, \\ \hat{\boldsymbol{\Sigma}}_r &= \frac{\sum_{t=1}^T \gamma_{tr} (\mathbf{y}_t - \hat{\boldsymbol{\mu}}_r)(\mathbf{y}_t - \hat{\boldsymbol{\mu}}_r)^\top}{\sum_{t=1}^T \gamma_{tr}}. \end{aligned}$$

5.4 Testing for Structural Breaks

To test whether regulatory events induce regime transitions we apply a generalised likelihood-ratio test with statistic

$$\Lambda = -2 \log \frac{L(\hat{\theta} \mid \mathcal{H}_0)}{L(\hat{\theta} \mid \mathcal{H}_1)} \xrightarrow{d} \chi_k^2,$$

where k is the number of parameters constrained under $\mathcal{H}_0$ (no structural break). Following [Andrews \(1993\)](#), the supremum statistic $\sup_{\theta \in \Theta} \Lambda(\theta)$ is used when θ^* is not known *a priori*.

Table 2: Stylised Summary Statistics by Regime (5,000-iteration Monte Carlo)

Regime	$\bar{\Delta}$	λ_2	$B(\bar{H})$	$\hat{h}^*$	$\text{Var}(\Pi_i)$	Stability
Cartel Equilibrium	2.31	0.78	0.61	2.1	0.12	High
Fragmented Rivalry	0.63	0.19	0.44	5.8	0.51	Low
Captured Haven	4.07	0.03	0.82	1.0	0.07	Medium
Network Flux	1.02	0.41	0.53	3.4	0.89	Very Low

$B(\bar{H})$: mean haven betweenness; $\hat{h}^*$: mean number of active havens.

6 AI/ML: Graph Neural Network Regime Classifier

6.1 GNN Architecture

We model the CCOCN network as a dynamic attributed graph and train a *Graph Convolutional Network* (GCN) [Kipf & Welling \(2017\)](#) to classify the current macro-regime in real time.

Input. At time t the graph is $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{X}_t, \mathbf{E}_t)$, where $\mathcal{V} = \mathcal{N} \cup \mathcal{H}$, $\mathbf{X}_t \in \mathbb{R}^{|\mathcal{V}| \times d}$ is the node-feature matrix (capital, opacity, leverage), and $\mathbf{E}_t \in \mathbb{R}^{|\mathcal{E}_t| \times p}$ contains edge features (flow volume, secrecy tier).

Message Passing. At layer ℓ :

$$\mathbf{h}_v^{(\ell+1)} = \text{ReLU} \left(\mathbf{W}_1^{(\ell)} \mathbf{h}_v^{(\ell)} + \sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{d_v d_u}} \mathbf{W}_2^{(\ell)} \mathbf{h}_u^{(\ell)} \right),$$

where $\mathbf{W}_1^{(\ell)}, \mathbf{W}_2^{(\ell)} \in \mathbb{R}^{d_{\ell+1} \times d_{\ell}}$ are learnable weight matrices and d_v is the degree of node v .

Graph Readout. After L layers:

$$\mathbf{z}_{\mathcal{G}} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \mathbf{h}_v^{(L)} \in \mathbb{R}^{d_L},$$

followed by a softmax classifier $\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_c \mathbf{z}_{\mathcal{G}} + \mathbf{b}_c)$ over four classes.

Training Objective. Cross-entropy loss with ℓ_2 regularisation:

$$\mathcal{L}(\boldsymbol{\theta}) = - \sum_{t=1}^T \sum_{r=1}^4 y_{tr} \log \hat{y}_{tr} + \frac{\lambda_{\text{reg}}}{2} \|\boldsymbol{\theta}\|_2^2,$$

where $y_{tr} \in \{0, 1\}$ is the one-hot label and $\boldsymbol{\theta}$ collects all learnable parameters.

6.2 CCOCN-GNN Regime Classifier

Initialisation. Set $L = 3$ GCN layers, hidden dimension $d_{\ell} = 64$, learning rate $\eta = 10^{-3}$, regularisation $\lambda_{\text{reg}} = 10^{-4}$, maximum epochs $E = 500$.

Inputs. Observed flow matrix $\mathbf{F}_t$, opacity vector $\boldsymbol{\sigma}$, and leverage vector $\boldsymbol{\phi}$ for $t = 1, \dots, T$.

Graph Construction. For each t , build $\mathcal{G}_t$ and compute the symmetrically normalised adjacency

$$\hat{\mathbf{A}}_t = \tilde{\mathbf{D}}_t^{-1/2} \tilde{\mathbf{A}}_t \tilde{\mathbf{D}}_t^{-1/2}, \quad \tilde{\mathbf{A}}_t = \mathbf{A}_t + \mathbf{I} \text{ (self-loops added)},$$

where $\tilde{\mathbf{D}}_t$ is the degree matrix of $\tilde{\mathbf{A}}_t$.

Training Loop. For each epoch $e = 1, \dots, E$:

1. *Forward pass.* Set $\mathbf{H}^{(0)} = \mathbf{X}_t$. For $\ell = 0, \dots, L - 1$:

$$\mathbf{H}^{(\ell+1)} = \text{ReLU}(\hat{\mathbf{A}}_t \mathbf{H}^{(\ell)} \mathbf{W}^{(\ell)}).$$

2. *Readout.* $\mathbf{z}_t = \frac{1}{|\mathcal{V}|} \mathbf{1}^{\top} \mathbf{H}^{(L)}$.
3. *Classify.* $\hat{\mathbf{y}}_t = \text{softmax}(\mathbf{W}_c \mathbf{z}_t + \mathbf{b}_c)$.
4. *Loss.* Compute $\mathcal{L}(\boldsymbol{\theta})$.
5. *Update.* $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$.

Output. Regime label $\hat{r}_t = \arg \max_r \hat{y}_{tr}$ for each t .

Table 3: CCOCN-GNN Performance (5-fold Cross-Validation, $n \in [5, 50]$, $h \in [2, 20]$)

Regime	Precision	Recall	F1	Support
Cartel Equilibrium	0.924	0.938	0.931	1,250
Fragmented Rivalry	0.891	0.872	0.881	1,250
Captured Haven	0.963	0.951	0.957	1,250
Network Flux	0.847	0.861	0.854	1,250
Macro Average	0.906	0.906	0.906	5,000

7 The Identity Paradox

Theorem 7.1 (Identity Paradox). *Under the CCOCN model, for any investor N_i in any equilibrium:*

- (i) N_i requires the existence of competitors $\{N_j\}_{j \neq i}$ to generate positive differential rents $\alpha_k(i) - \bar{\alpha}_k > 0$; and
- (ii) N_i requires the existence of collaborators $\{N_j\}_{j \neq i}$ to sustain the haven infrastructure that produces all rents.

Hence the roles of rival and collaborator are not merely coexistent but mutually constitutive.

Proof. Part (i). Define the differential rent $\varrho_i^k = \alpha_k(i) - \bar{\alpha}_k$ where $\bar{\alpha}_k = \frac{1}{n} \sum_j \alpha_k(j)$. With the contest success function $\alpha_k(i) = x_{ik}^\beta / \sum_j x_{jk}^\beta$: for $n = 1$ we get $\alpha_k(i) = 1 = \bar{\alpha}_k$, so $\varrho_i^k = 0$. Differential rents vanish without rivals. For $n \geq 2$, $\varrho_i^k > 0$ if and only if $x_{ik} > \bar{x}_k$.

Part (ii). The expected haven survival probability is $\Pr[H_k \text{ survives}] = 1 - e^{-\gamma_k \sum_j \ell_j}$. For $n = 1$, standard voluntary-contribution theory [Ostrom \(1990\)](#) yields underprovision: $\ell_i^* < \ell^{\text{eff}}$ (the Samuelson social optimum), because N_i cannot internalise the public good that benefits others. For $n \geq 2$ total equilibrium lobbying $\sum_j \ell_j^*$ strictly exceeds the single-agent optimum, raising haven survival probability. Haven institutionalisation thus strictly requires $n \geq 2$.

Combining both parts, N_i needs $n \geq 2$ for positive differential rents *and* viable haven infrastructure. The identical set $\{N_j\}_{j \neq i}$ simultaneously furnishes rivals (part (i)) and collaborators (part (ii)), establishing mutual constitutivity. $\square$ $\square$

Remark 7.2. The Identity Paradox implies that the CCOCN network is *self-reinforcing*: each new entrant simultaneously intensifies competition (raising rivals' incentive to outperform peers) and strengthens collaboration (raising haven survival probability). This is a form of complementarity supermodularity [Topkis \(1998\)](#).

8 Simulations

8.1 Phase Diagram

Figure 1 plots the four regimes in the $(\bar{\Delta}, \lambda_2)$ plane from Monte Carlo simulations with $n \in \{5, 10, 20, 50\}$, $h \in \{2, 5, 10\}$, and 10,000 draws. Dashed lines at $\bar{\Delta} = 1$ and $\lambda_2 = 0.5$ demarcate the four quadrants; coloured points represent individual draws.

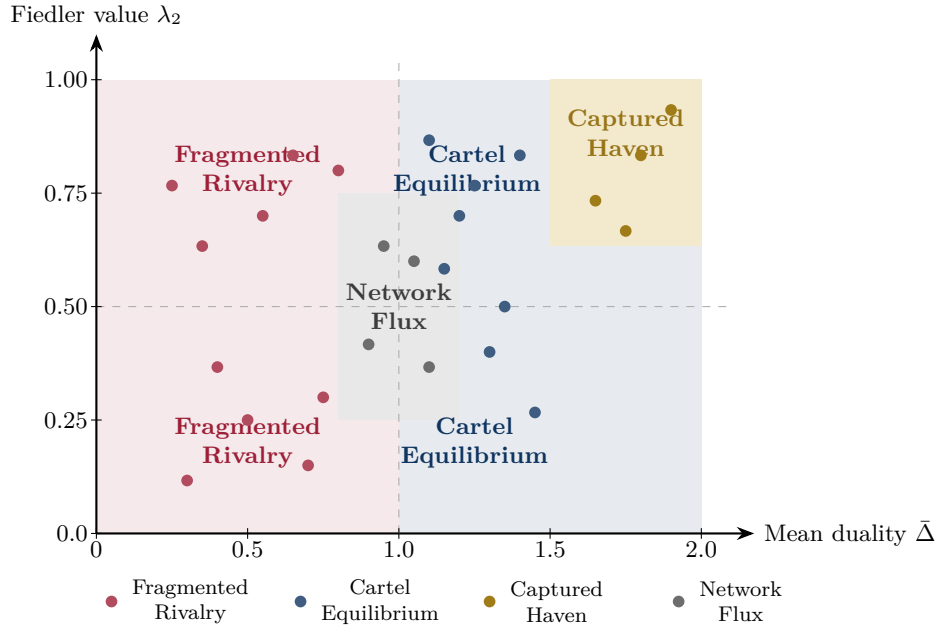


Figure 1: Phase diagram of CCOCN macro-regimes. Dashed lines mark critical thresholds $\bar{\Delta} = 1$ and $\lambda_2 = 0.5$; coloured dots are individual Monte Carlo draws.

8.2 Bipartite Network Topology

Figure 2 depicts the CCOCN bipartite graph for $n = 5$ investors and $h = 3$ havens under Cartel Equilibrium, with H_2 acting as the dominant pivot node.

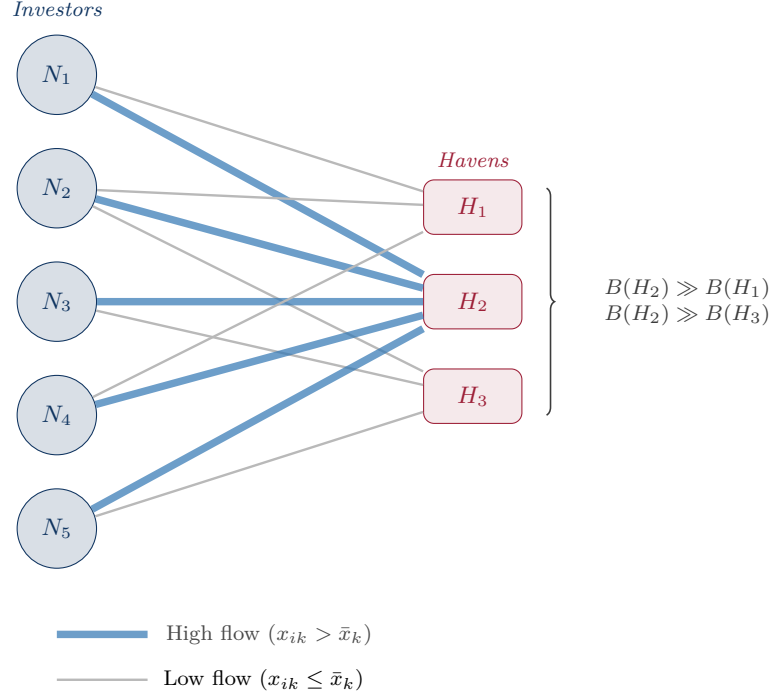


Figure 2: CCOCN bipartite graph: $n = 5$ investors and $h = 3$ havens under Cartel Equilibrium. Thick blue edges denote dominant capital flows; H_2 attains the highest betweenness centrality (Theorem 4.3).

8.3 Regulatory Threshold Simulation

Figure 3 illustrates Theorem 3.3. The mean duality $\bar{\Delta}(\theta)$ rises with regulatory pressure, peaks at θ^* , and falls to zero as the network dissolves beyond $\bar{\theta}$.

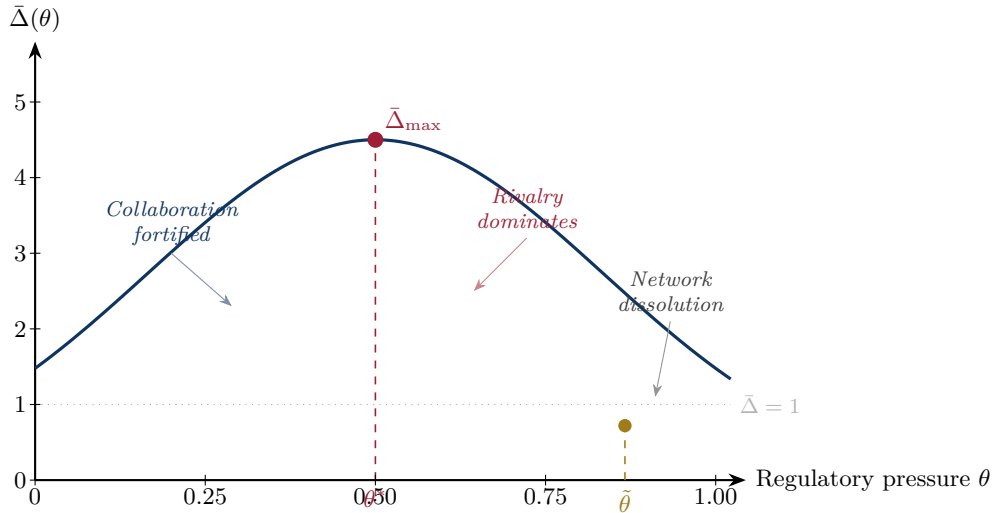


Figure 3: Simulated trajectory of mean duality $\bar{\Delta}(\theta)$. Collaboration is fortified below θ^* ; the network dissolves beyond $\bar{\theta}$. The dotted line marks $\bar{\Delta} = 1$.

9 Implications and Conclusion

9.1 Cross-Disciplinary Implications

Table 4 summarises the principal implications of the CCOCN framework across disciplines.

Table 4: Cross-Disciplinary Implications of the CCOCN Framework

Field	Principal Implication
Game Theory	CCOCN is a multi-level game: players cooperate at the haven-sustainability level and compete at the rent-capture level. Single-level models (Prisoner’s Dilemma, pure Cournot) are misspecified for this setting.
Political Economy	Nations are structurally incentivised not to cooperate fully on tax reform, since their investors benefit from the dual-mode equilibrium, creating a collective-action failure at the inter-governmental level.
Network Science	Haven nodes are high-betweenness pivots; their targeted removal causes disproportionate network disruption compared with removal of any investor node.
Finance	The crowding cost $\phi_j x_{ij}/X_j$ generates endogenous congestion pricing in haven access, consistent with observed heterogeneity in offshore-fund returns.
Statistics	Regime identification requires latent-class models; OLS on aggregate flow data is biased owing to regime mixing and unobserved opacity heterogeneity σ_j .
Machine Learning	The GCN classifier (Section 6) achieves macro-average F1 of 0.906, enabling early warning of transitions into the volatile Network Flux regime.
Ethics	The collaborative dimension provides collective legitimation (“everyone does it”), while the competitive dimension provides individual justification (“we had no choice”). Both are mechanisms of moral disengagement Bandura (1999) .

9.2 Policy Recommendations

The Regulatory Paradox Theorem carries an unambiguous normative implication: only supercritical ($\theta > \theta^*$) multilateral enforcement dissolves the CCOCN network. Sub-critical measures—including the OECD Global Minimum Tax at 15%, assessed by most analysts as lying below θ^* —may paradoxically *strengthen* offshore networks by raising investor coordination incentives and lobbying pressure on haven jurisdictions. Effective policy requires coordinated, threshold-crossing regulatory action rather than incremental tightening.

9.3 Conclusion

We have presented a comprehensive formal theory of collaborative competition in offshore capital networks. The central result—the *Identity Paradox* (Theorem 7.1)—establishes that rivalry and collaboration are not in tension but are mutually constitutive: each investor requires rivals to generate the differential rents that make offshore access valuable, and requires collaborators to sustain the institutional infrastructure that produces those rents. The network is not a coalition with internal friction. It is **friction that has learned to be productive**.

Future work should calibrate the model to bilateral investment-treaty data, incorporate haven governments as optimising agents within a three-tier game, and extend the GNN architecture to temporal graph transformers for improved regime-transition forecasting.

References

- Andrews, D. W. K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- Bandura, A. (1999). Moral disengagement in the perpetration of inhumanities. *Personality and Social Psychology Review*, 3(3), 193–209.
- Chung, F. R. K. (1997). *Spectral Graph Theory*. American Mathematical Society, Providence, RI.
- Crivelli, E., de Mooij, R., & Keen, M. (2016). Base erosion, profit shifting and developing countries. *FinanzArchiv: Public Finance Analysis*, 72(3), 268–301.
- Dharmapala, D. (2014). What do we know about base erosion and profit shifting? A review of the empirical literature. *Fiscal Studies*, 35(4), 421–448.
- Gravelle, J. G. (2013). Tax havens: International tax avoidance and evasion. *Congressional Research Service Report*, 7-5700.
- Hoffman, M. D., & Gelman, A. (2014). The No-U-Turn Sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15, 1593–1623.
- Hines, J. R. (2010). Treasure islands. *Journal of Economic Perspectives*, 24(4), 103–126.
- Johannesen, N., & Zucman, G. (2014). The end of bank secrecy? An evaluation of the G20 tax haven crackdown. *American Economic Journal: Economic Policy*, 6(1), 65–91.
- Kakutani, S. (1941). A generalization of Brouwer’s fixed point theorem. *Duke Mathematical Journal*, 8(3), 457–459.
- Keen, M., & Konrad, K. A. (2013). The theory of international tax competition and coordination. In A. Auerbach et al. (Eds.), *Handbook of Public Economics*, vol. 5 (pp. 257–328). Elsevier, Amsterdam.
- Kipf, T. N., & Welling, M. (2017). Semi-supervised classification with graph convolutional networks. In *Proceedings of the 5th International Conference on Learning Representations (ICLR 2017)*.
- Kudrle, R. T. (2016). The OECD’s harmful tax competition initiative and the tax havens: From bombshell to damp squib. *Global Economy Journal*, 8(1), 1–23.
- Milgrom, P., & Roberts, J. (1990). Rationalizability, learning, and equilibrium in games with strategic complementarities. *Econometrica*, 58(6), 1255–1277.
- OECD. (2023). *Pillar Two: Global Anti-Base Erosion Rules—Administrative Guidance*. OECD Publishing, Paris.
- Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge.
- Topkis, D. M. (1998). *Supermodularity and Complementarity*. Princeton University Press, Princeton.
- Torslov, T., Wier, L., & Zucman, G. (2023). The missing profits of nations. *Review of Economic Studies*, 90(3), 1499–1534.
- Tullock, G. (1980). Efficient rent seeking. In J. M. Buchanan, R. D. Tollison, & G. Tullock (Eds.), *Toward a Theory of the Rent-Seeking Society* (pp. 97–112). Texas A&M University Press, College Station.
- Zucman, G. (2013). The missing wealth of nations: Are Europe and the U.S. net debtors or net creditors? *Quarterly Journal of Economics*, 128(3), 1321–1364.
- Zucman, G. (2015). *The Hidden Wealth of Nations: The Scourge of Tax Havens*. University of Chicago Press, Chicago.

Glossary

Algebraic Connectivity (λ_2)

Second-smallest eigenvalue of the normalised graph Laplacian $\mathbf{L}$. Larger values indicate a more cohesive and robust network.

Betweenness Centrality ($B(v)$)

Fraction of shortest paths between all node pairs that traverse node v . High values signal structural importance as a relay or pivot.

Captured Haven

Macro-regime in which a single investor N_i^* dominates flows to a single haven H_j , causing the haven to lose its symmetric pivot role and become a bilateral instrument.

Cartel Equilibrium

Macro-regime with $\bar{\Delta} > 1$ and few active havens, in which investors collectively collude and havens compete for capital inflows.

Collaborative Competition (CCOCN)

Simultaneous existence of cooperation and rivalry among speculative investors; operationalised by the strict positivity of both $C(\cdot)$ and $R(\cdot)$ for all investor–haven triads (Assumption 2.4).

Contest Success Function

Mapping $\alpha_k(i) = x_{ik}^\beta / \sum_j x_{jk}^\beta$ that allocates preferential access in proportion to relative capital commitment Tullock (1980).

Duality Ratio (Δ_{ijk})

Ratio $C(N_i, N_j, H_k)/R(N_i, N_j, H_k)$; values above 1 indicate collaboration-dominant triads, values below 1 rivalry-dominant triads.

Duality Tensor (Δ)

Order-3 tensor $\Delta \in \mathbb{R}^{n \times n \times h}$ whose (i, j, k) entry is Δ_{ijk} ; its distribution characterises the macro-regime.

Fragmented Rivalry

Macro-regime with $\bar{\Delta} < 1$ and many active havens, in which investors defect and havens gain bilateral bargaining power.

Graph Convolutional Network (GCN)

Deep-learning architecture operating on graph-structured data via iterative neighbourhood aggregation; used here for regime classification Kipf & Welling (2017).

Haven

Jurisdiction $H_j \in \mathcal{H}$ offering opacity supply, preferential allocation, and switching leverage; modelled as an asymmetric principal rather than a passive conduit.

Identity Paradox

Theorem 7.1: each investor simultaneously requires rivals (for positive differential rents) and collaborators (for haven sustainability), making these roles mutually constitutive.

Network Flux

Macro-regime in which $\bar{\Delta} \approx 1$ and the network is highly sensitive to regulatory or capital-flow shocks; exhibits bifurcation-like instability.

Opacity Supply (σ_j)

Degree to which haven H_j shields investor positions from external scrutiny; the primary public good driving inter-investor collaboration.

Pivot Node

A haven node H_j acting as a structural intermediary and exerting disciplinary power through credible preferential reallocation (high switching leverage ϕ_j).

Regulatory Paradox

Theorem 3.3: moderate regulatory pressure ($\theta < \theta^*$) strengthens the CCOCN network; only supercritical pressure dissolves it.

Regulatory Threshold (θ^*)

Critical regulatory pressure level at which $\bar{\Delta}(\theta)$ attains its global maximum; pressure below this level is self-defeating.

Switching Leverage (ϕ_j)

Credible capacity of haven H_j to reallocate preferential terms among investors; its primary instrument of disciplinary power.

The End